Scenario: "Design the Component Architecture for This Feature"

10.2 - Scenario: "Design the Component Architecture for This Feature"

🎯 Why This Question Exists

This tests your ability to think about component structure before writing code.

Interviewers want to see if you can break a feature into well-organized, reusable pieces.

The Framework: A 4-Step Process

Example: "Design the component architecture for an e-commerce product listing page"

Step 1: Identify the major sections of the UI

- Header with search bar and cart icon
- Filter sidebar (category, price range, rating)
- Product grid (product cards with image, name, price, add-to-cart)
- Pagination or infinite scroll
- Sort dropdown (price, popularity, newest)

Step 2: Define the component tree

JAVASCRIPT

Copy

```
ProductListingPage
  SearchBar
  FilterSidebar
    CategoryFilter
    PriceRangeFilter
    RatingFilter
  SortDropdown
  ProductGrid
    ProductCard (repeated)
      ProductImage
      ProductInfo
      AddToCartButton
  Pagination
```

Step 3: Determine where state lives

- **Search query, active filters, sort order, current page:** in ProductListingPage (affects API call)
- **Cart items:** in a CartContext (global, used by Header and AddToCartButton)
- **Product data:** fetched via a `useProducts` custom hook that combines filters + sort + page

Step 4: Identify data flow and communication

- FilterSidebar calls `onFilterChange` callback to update parent state
- ProductCard calls `onAddToCart` which dispatches to CartContext
- URL params stay in sync with filters for shareable links (using useSearchParams)

Sample Answer

the product grid, and pagination. Then I'd build a component tree with ProductListingPage as the parent owning the filter, sort, and page state. FilterSidebar and SortDropdown communicate back to the parent through callbacks. ProductGrid receives the filtered products as props. Cart state lives in a CartContext so both the header's cart icon and the AddToCartButton can access it. I'd extract the API call into a useProducts custom hook and sync filter state with URL params so users can share filtered views."

QUICK CHECK — no marks, just for you.

1. In the product listing architecture described in the lesson, which component owns the search query, active filters, sort order, and current page state?

- ○ FilterSidebar, since it renders the filter controls that change them
- ○ CartContext, so that every component in the tree can read them
- ○ ProductListingPage, the parent of the sidebar, grid, and pagination
- ○ ProductGrid, since it is the component that displays the results

Check answer

2. According to the lesson, why is cart state kept in a CartContext instead of being held in ProductListingPage like the filter state?

- ○ Because context re-renders are always cheaper than passing props down a tree
- ○ Because state that changes frequently cannot legally live in a page-level component
- ○ Because both the header's cart icon and AddToCartButton need it, so it is treated as global state

Because cart data must be fetched from the server and only context can hold async data

Check answer

3. The lesson recommends keeping URL params in sync with the filter state. What benefit does it give for doing this?

- ◯ Users get shareable links to a filtered view
- ◯ It lets FilterSidebar update state without any callback to the parent
- ◯ It prevents ProductCard from re-rendering when filters change
- ◯ It removes the need for the useProducts custom hook

Check answer

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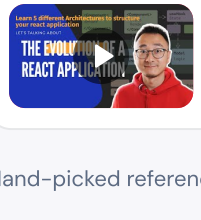
ARTICLES & DOCS

Thinking in React
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Structure your React app: small → large
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Hand-picked references for this lesson.